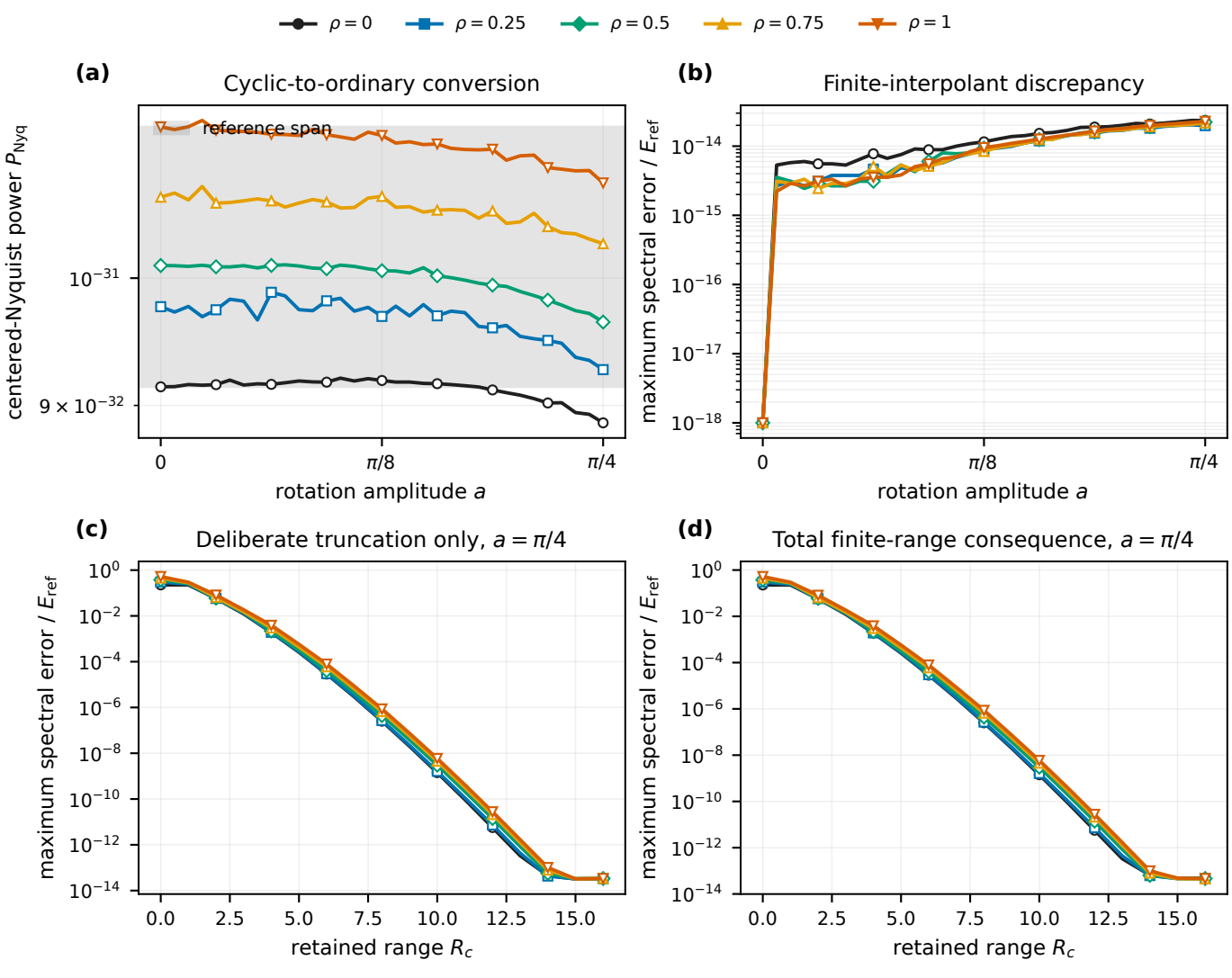
ρ = 0
ρ = 0.25
ρ = 0.5
ρ = 0.75
ρ = 1
(a)
Cyclic-to-ordinary conversion
reference span
centered-Nyquist power P_{Nyq}
10^{-31}
9×10^{-32}
0
π/8
π/4
rotation amplitude a
(b)
Finite-interpolant discrepancy
maximum spectral error / E_{ref}
10^{-14}
10^{-15}
10^{-16}
10^{-17}
10^{-18}
0
π/8
π/4
rotation amplitude a
(c)
Deliberate truncation only, $a = \pi/4$
maximum spectral error / E_{ref}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
0.0
2.5
5.0
7.5
10.0
12.5
15.0
retained range R_c
(d)
Total finite-range consequence, $a = \pi/4$
maximum spectral error / E_{ref}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
0.0
2.5
5.0
7.5
10.0
12.5
15.0
retained range R_c